MARITIME AI INITIATIVE

AI-Powered Fuel & Emissions Optimization

A proof-of-concept business case for turning predictive AI into funded fleet operations initiatives

Prepared for Executive Leadership Review

EXECUTIVE SUMMARY

From volatile fuel spend to a data-driven cost advantage

Goal

Build an AI-powered system that optimizes fuel use and emissions across the fleet.

Objective

Deliver cost-effective, environmentally compliant, and reliable vessel operations.

Approach

Data-driven AI/ML models for prediction, optimization, and anomaly detection.

Outcome

A working proof of concept that identifies fundable interventions with clear savings.

Bottom line: A working model already points to fuel savings in the 5-15% range on speed optimization alone — the next step is converting this proof of concept into funded fleet initiatives.

THE BUSINESS CHALLENGE

Four forces are squeezing margin and raising compliance risk

High & volatile fuel costs	Fuel represents 30-50% of operating expense and swings with global markets.
Stringent environmental regulation	IMO 2020 and EEXI/CII compliance carry real financial and reputational stakes.
Operational inefficiency	Suboptimal routes, speeds, and trim quietly erode margin voyage after voyage.
Data scarcity & fragmentation	Vessel, voyage, and weather data sit in disconnected systems, limiting visibility.

DATA FOUNDATION

Six data streams feed a single operating picture

Vessel Operational

RPM, fuel flow, speed, draft sensors

Voyage & Route

Planned vs. actual routing, cargo info

Environmental

Wind, waves, currents, weather

Fuel

Type, consumption, cost

Vessel Static

Specifications, maintenance history

Emissions

CO_2, SO_x, NO_x output

Three AI/ML capabilities, one integrated system

Predict

SUPERVISED LEARNING

Ensemble methods (Random Forest, XGBoost) and LSTMs forecast fuel consumption and emissions from operating conditions.

Optimize

REINFORCEMENT LEARNING

Reinforcement learning and metaheuristics recommend optimal speed and routing under real-time constraints.

Detect

ANOMALY DETECTION

Isolation Forest, autoencoders, and clustering flag equipment and performance anomalies early.

PROOF OF CONCEPT

The model already works — and tells us where value is hiding

0.56

R^2 — predictive fit
(target 0.70-0.90+ at scale)

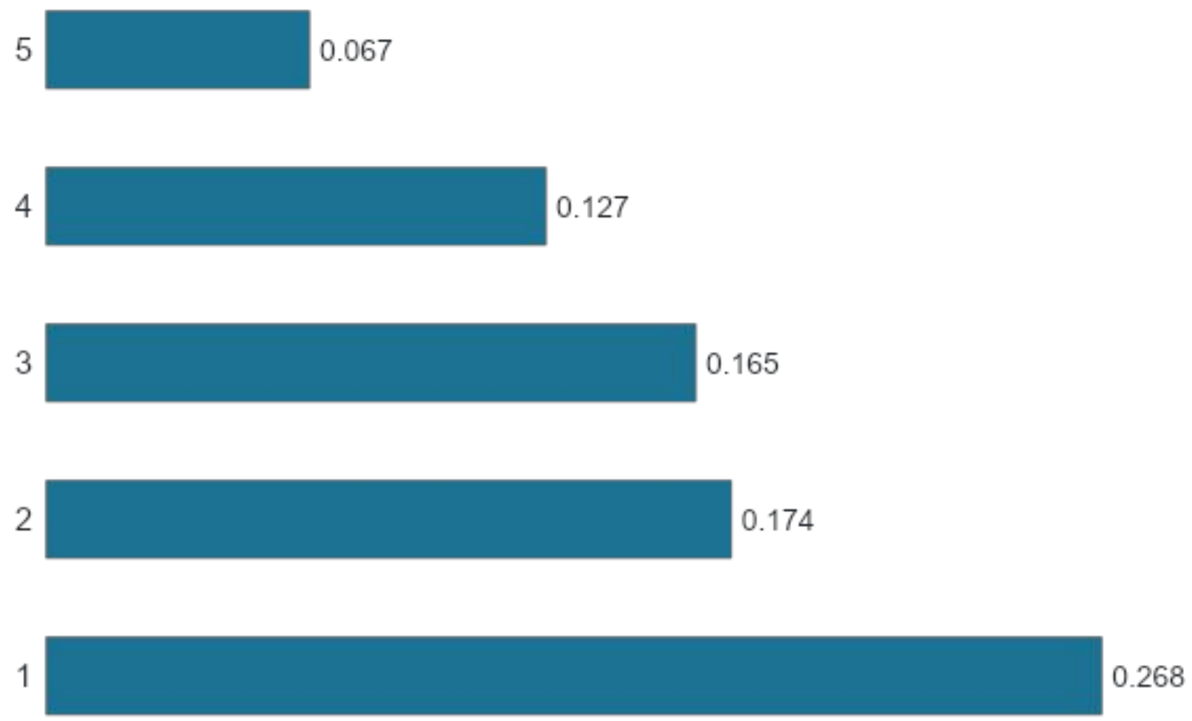
19.5 kg/h

Mean absolute error
(~6.5% of typical consumption)

1,000

Synthetic vessel-voyage
records used to validate approach

What drives fuel consumption



WHAT THE MODEL TELLS US

Four levers leadership can act on today

Speed

Small speed reductions yield outsized fuel savings — the single strongest lever available.

Weather

Wave height and wind speed are major drivers — proactive routing is a high-value capability.

Cargo

Load distribution affects resistance — optimizing stowage reduces fuel burn.

Engine health

RPM patterns flag inefficiency early — a bridge to predictive maintenance.

FROM MODEL TO MONEY

Five interventions, ranked by expected fuel savings

Dynamic Speed Optimization

AI-recommended optimal speed balancing ETA, fuel & conditions

Weather Routing & Avoidance

Routes that avoid high waves, headwinds, adverse currents

Engine Performance Monitoring

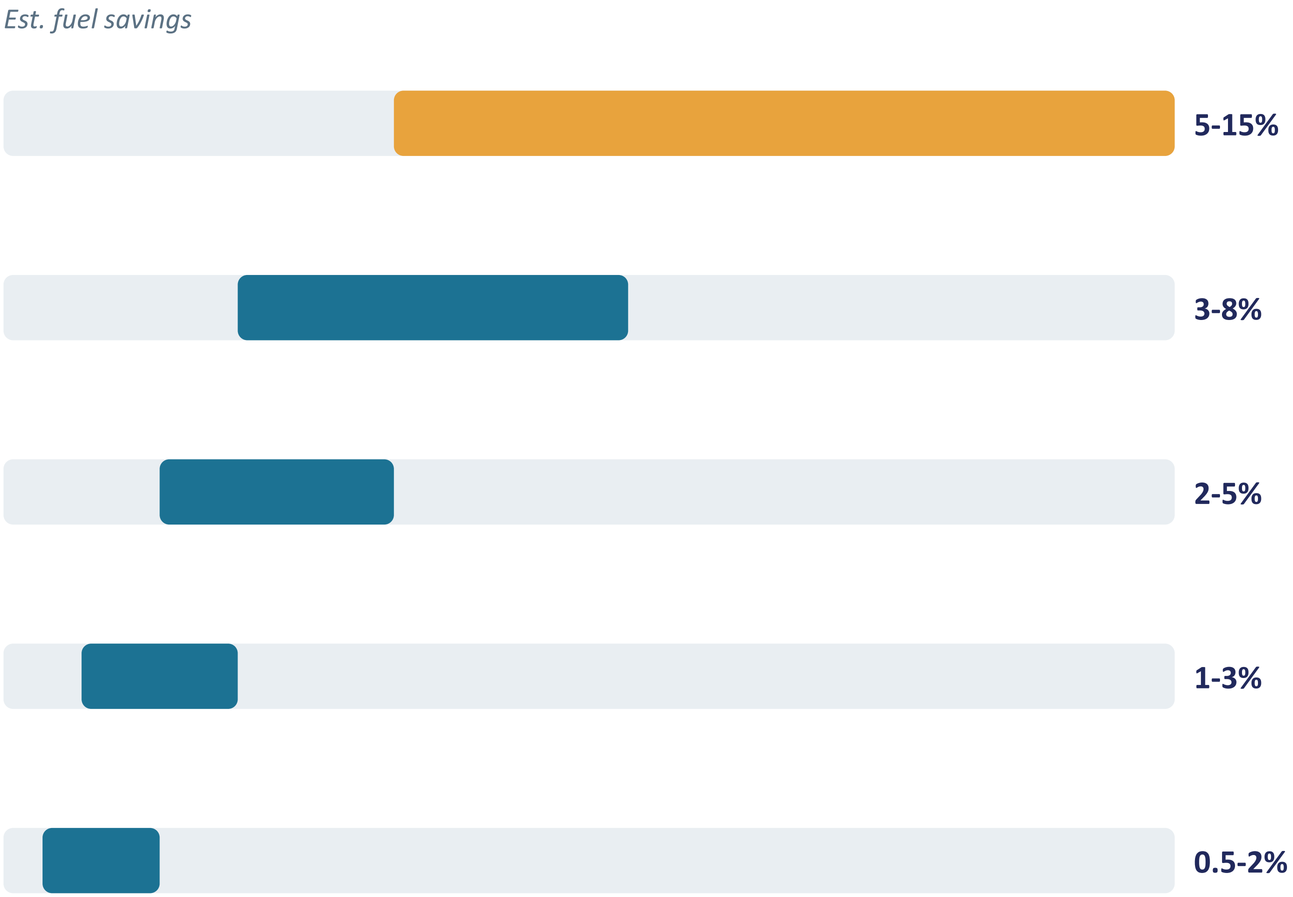
Real-time deviation detection enabling proactive maintenance

Cargo Load Optimization

Optimal loading configuration to reduce hull resistance

Trim & Draft Optimization

Real-time ballast recommendations for hydrodynamic efficiency



THE ASK

Converting this proof of concept into three funded initiatives

Phase 1 • 0-3 months

Data Foundation

Stand up automated pipelines connecting vessel sensors, voyage, and weather data. Validate data quality end to end.

Phase 2 • 3-7 months

Model Scale-Up

Retrain on real fleet data, lift R^2 toward 0.70-0.90+, and pilot dynamic speed & weather routing on select vessels.

Phase 3 • 7-12 months

Fleet Rollout

Deploy dashboards fleet-wide, integrate with voyage management systems, and institutionalize through crew training.

WHAT SUCCESS REQUIRES

Five conditions for this to scale beyond a pilot

Data quality & integration

Robust pipelines and automated validation across vessel and shore systems.

Model refinement & XAI

Continuous learning, A/B testing, and explainability to build crew and officer trust.

Intuitive interfaces

Onboard and shore dashboards integrated with existing voyage management systems.

Change management

Structured training and adoption plan for crew and shore management.

Scalable governance

Secure, well-governed architecture with clear KPIs and agile delivery.

DECISION REQUESTED

Approve Phase 1 to move from proof of concept to a funded fleet initiative

Sponsor a data pipeline pilot

2-3 vessels, 90 days, to validate real fleet data quality

Allocate scale-up budget

For model retraining and dashboard integration in Phase 2

Name an executive sponsor

To champion change management and crew adoption